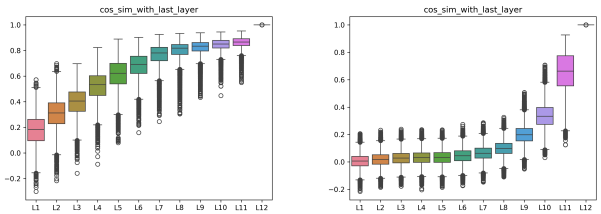
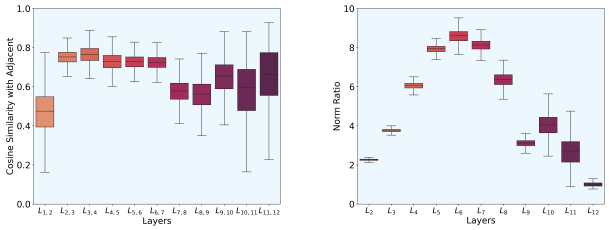


(a) CIFAR10 (b) IN1K

Figure 10: Sample-wise cosine similarity of features from shallow layers and the last-hidden layer. The DeiT-S model is trained with standard training on CIFAR-10 and ImageNet. It shows there are rare samples with negative sample-wise cosine similarity.



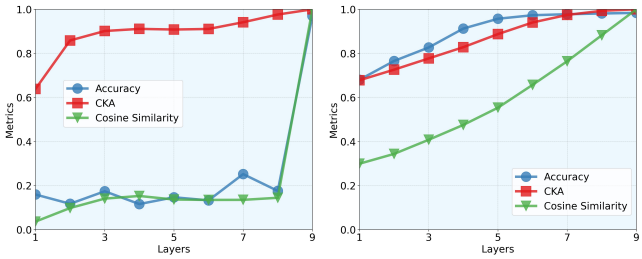
(a) COS of adjacent layers (b) Norm Ratio

Figure 11: Cosine similarity of features from adjacent layers $\mathrm{COS}(\boldsymbol{h}^{(\ell-1)}, \boldsymbol{h}^{(\ell)})$ and norm ratios $\|\boldsymbol{h}^{(\ell)}\|/||f_{\theta^{(\ell)}}(\boldsymbol{h}^{(\ell)})||$ distributions. The DeiT-Small model is trained on Imagenet-1K and evaluated on its validation dataset.

Most transformer architectures include skip connections, which are added after the (i) self-attention layer and (ii) MLP layer. According to residual transformation (1), we obtain,

$$\begin{aligned}\boldsymbol{H}^{(\ell+1)} &= \mathrm{MLP}(\mathrm{LN}(\mathrm{MSA}(\mathrm{LN}(\boldsymbol{H}^{(\ell)}) + \boldsymbol{H}^{(\ell)})) + \mathrm{MSA}(\mathrm{LN}(\boldsymbol{H}^{(\ell)})) + \boldsymbol{H}^{(\ell)} \\ &= \boldsymbol{H}^{(\ell)} + f_{\theta^{(\ell)}}(\boldsymbol{H}^{(\ell)})\end{aligned}$$

And the feature $\boldsymbol{h}^{(\ell)}$ is a special token of $\boldsymbol{H}^{(\ell)}$. To investigate the effect of residual connections, we calculate the norm ratio $\|\boldsymbol{h}^{(\ell)}\|/||f_{\theta^{(\ell)}}(\boldsymbol{h}^{(\ell)})||$, which is the ratio of the norm of skip connection $\boldsymbol{h}^{(\ell)}$ to the norm of the long branch $f_{\theta^{(\ell)}}(\boldsymbol{h}^{(\ell)})$. The results are displayed in Figure 11. High norm ratios suggest that skip connections significantly influence the representational structure of ViT.



(a) Without residual connections (b) With residual connections

Figure 12: Comparison of layerwise accuracy, COS(COsine Similarity), and CKA (Centered Kernel Alignment) with the last layer of the 9-layer MLP models with and without residual connection on the MNIST validation dataset. The models are trained from scratch using standard training. In the left figure, CKA fails to accurately reflect the change in layerwise accuracy for the MLP without residual connection. In the right figure, the presence of a residual connection is the reason why CKA works well, as it helps eliminate rotation ambiguity.

To provide further evidence that residual connections resolve the rotation ambiguity, we compared the MLP model with and without these connections and computed their COS and CKA values. For the MLP model without residual connections, as shown in Figure 12(a), the CKA value is not consistent with accuracy and cosine similarity. A high CKA value might indicate significant similarity between features across layers, but it does not necessarily correlate with high classification accuracy. This inconsistency primarily results from the fact that CKA does not account for rotation in the feature space, suggesting that features could rotate without the residual connections. In contrast, for the MLP model with residual connections, as depicted in Figure 12(b), the CKA value aligns with layerwise accuracy, indicating that residual connections effectively eliminate the rotation ambiguity of features.

B.3 Aligned Training Details

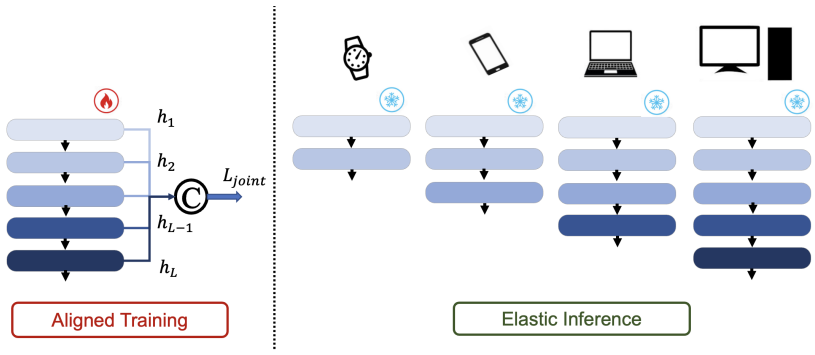


Figure 13: Aligned training of transformer using joint CE loss of all layer features with common classifier and elastic inference for different memory constrains. Once the model is trained using the aligned method, it can fit all devices. Features from darker layers indicate better performance.

Illustration of Train Once and Fit all devices. Figure 13 illustrate how aligned training support train once and fit all devices. After aligned training, one can directly fetch from shallow to deep layers of transformer according to the device computational resources and memory constrains.

Alternative Approach for Enhancing Layer-wise Representation Similarity. In addition to using aligned training loss to enhance similarity, another method is to add the cosine similarity as a regularization term to the loss function.

$$\mathcal{L}_{\text{sim}}(\boldsymbol{h}^{(\ell)}, \boldsymbol{h}^{(L)}) = \sum_{l=1}^{L} \lambda_\ell (1 - \cos(\boldsymbol{h}^{(\ell)}, \boldsymbol{h}^{(L)}))$$

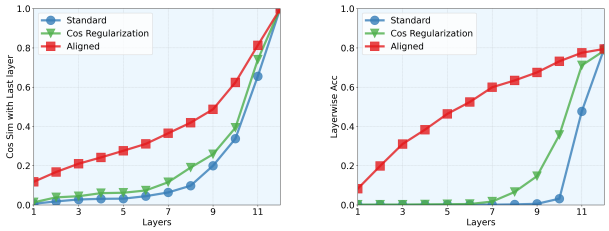
And the total loss is the sum of this two term:

$$\mathcal{L}_{\text{CE-reg}}(\boldsymbol{x}, y) = \mathcal{L}_{\text{CE}}(\boldsymbol{W}\boldsymbol{h}^{(L)}, y) + \beta \mathcal{L}_{\text{sim}}(\boldsymbol{h}^{(\ell)}, \boldsymbol{h}^{(L)})$$

where $\beta > 0$ is the regularization coefficient. According to Figure 14, the regularization term contributes minimally to the improvement of cosine similarity and layer-wise accuracy, compared to aligned training methods.

Using the CE-reg loss results in poor layerwise accuracy and lower cosine similarity compared to the aligned loss. The likely reason for this is an imbalance between the COS alignment objective and the primary classification objective. Our intuition is that directly optimizing for high COS alignment may fail because the COS alignment loss primarily focuses on aligning features across layers, without necessarily making the features discriminative enough for the classification task. In contrast, the cross-entropy (CE) loss directly optimizes for classification, and as a consequence, it naturally improves COS alignment. This suggests that while COS alignment is important, it may not be sufficient on its own without the robust guidance provided by the CE loss.

Another approach involves adding the CKA term as a regularization term to the CE loss. However, this approach may not be effective and has several drawbacks. First, CKA is not always reliable in



(a) Cos Similarity (b) Layerwise Acc

Figure 14: Cosine similarity with last layer and layerwise accuracy of standard training using $\mathcal{L}_{\text{CE}}(\boldsymbol{W}\boldsymbol{h}^{(\ell)}, y)$, standard-Reg training using $\mathcal{L}_{\text{CE-reg}}(\boldsymbol{x}, y)$ and Aligned training using $\mathcal{L}_{\text{aligned}}(\boldsymbol{x}, y)$. The 12 layers DeiT model is trained on ImageNet1K dataset. Regularization term helps little for improving the cosine similarity and layerwise accuracy. But our aligned training improves both a lot.

representing layer-wise similarity across all settings, particularly in scenarios with rotation ambiguity or when residual connections are absent. Second, it is computationally expensive, as it requires computing the Gram matrix to evaluate relationships between features. Lastly, CKA might not perform better than the COS regularization term and may yield similar results, falling short compared to the aligned loss. In transformers, both COS and CKA measure feature similarity and tend to exhibit similar trends. As shown in Figure 14, the COS regularization term contributes minimally to improving cosine similarity and layer-wise accuracy. Based on this, we infer that using CKA as a regularization term would similarly have a minimal impact on enhancing these metrics. Therefore, aligned training approaches may be more effective than relying solely on regularization terms.

Setup for Aligned Training in ViT. It's reported [30] that training only with this aligned loss would cause the performance drop in the last layer. So following [30], we choose the "alternating" training approach, which alternates objectives based on the iteration number. During odd-numbered iterations, we use the CE loss of the final layer $\mathcal{L}_{\text{CE}}(\boldsymbol{W}\boldsymbol{h}_L, y)$. For even-numbered iterations, the strategy involves using the aligned loss $\mathcal{L}_{\text{aligned}}(\boldsymbol{x}, y)$.

Note that this training strategy, which uses a common classifier, no longer requires the KL-divergence term that is commonly used in mutli-exit/classifeirs training. This is because the deep layers have been trained to capture the abstract and discriminative features of the input data, effectively serving as the teacher model. The KL-divergence term is typically used to guide the shallower layers. However, when we use a common classifier, our aligned training method becomes a latent knowledge self-distillation method. The shallow layers can mimic or align their feature representations with those of the deep layers by aligning with the common classifier. As such, the deep layers, with their advanced feature representations, act as the teachers, while the shallow layers, in their quest to improve their feature extraction capabilities, assume the role of students. Therefore, the KL-divergence term is no longer necessary.

Linear Increasing Weight vs. Uniform Weight for Loss. In (5), we define the layer weights to increase linearly as $\lambda_\ell = 2\ell/(L(L+1))$. For the ablation study, we also consider a uniform weighting strategy, where all layers are assigned the same weight, i.e., $\lambda_\ell = 1/L$. As shown in Figure 15, using uniform weights in the loss function tends to improve the performance of shallow layers but degrade the final layer. This occurs because shallow layers, which typically learn general but less informative features, produce larger losses, while deeper layers achieve smaller losses. Uniformly weighting all layers disproportionately emphasizes shallow layers and diminishes the importance of deeper ones. In contrast, linearly increasing the weights places greater emphasis on deeper layers, resulting in superior accuracy for the final layer. Therefore, linear increasing weight is selected instead of uniform weight for aligned training.

Aligned Training Enhance Neural Collapse. The aligned training method aligns features with the last-layer classifier from shallow to deep layers, enhancing the neural collapse ($\mathcal{NC}$) phenomenon across layers. As shown in Figure 16, aligned training promotes progressive compression